

Philips GmbH Market DACH Health Systems
Recruiting Team
Röntgenstraße 22
22335 Hamburg
Germany

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Application for Research Scientist – AI in Computed Tomography Image Formation

Dear Philips Recruiting Team,

I am applying for the Research Scientist – AI in Computed Tomography Image Formation position because I am strongly interested in the technical challenge of reconstructing diagnostically useful CT images from noisy, dose-constrained measurements, and in Philips' mission to make high-quality healthcare accessible to everyone. I believe my experience developing and validating deep-learning methods for medical imaging across various modalities, together with my work with clinical and research partners, would enable me to contribute effectively to Philips' CT image formation team.

My interest in clinically useful machine learning began during my studies through projects conducted with a local hospital. There, I developed a semantic-segmentation model for automated tissue-density analysis of 3D lung CT scans. Working with doctors showed me how strongly the value of a model depends on the clinical question it addresses and on close collaboration throughout its development. I continued in this direction with my thesis on anatomical landmark localization for orthopedics. In cooperation with spine specialists, I improved the previous model by 30%; the work was presented at DWG 2023.

Since then, I have broadened this experience across further medical-imaging problems. In a three-year BMBF research project, I developed 3D instance-segmentation and image-registration methods for anisotropic light-sheet microscopy images in liver-cancer research. At RAYLYTIC, I later developed, validated, and deployed an end-to-end 2D–3D registration system for estimating implant pose from X-rays. The work improved the system's acceptance rate by 45% and reduced error by 15%, was presented at EFORT 2026, and is now used at scale in current clinical trials. These projects required systematic experimentation, quantitative validation, and communication across technical and clinical disciplines.

I am particularly interested in applying this background to learned reconstruction and denoising for conventional, spectral, and photon-counting CT. The interaction between spatial and temporal resolution, noise, artifacts, and dose makes this a technically demanding research area with a direct clinical purpose. I would bring hands-on experience with convolutional models, volumetric data, registration, segmentation, Python and C++, as well as the discipline to carry methods from initial experiments through validation and dependable implementation.

Philips' emphasis on patient safety, quality, and clinically meaningful innovation strongly matches how I want my work to be used. I would welcome the opportunity to discuss how my medical-imaging research and engineering experience could contribute to better CT image formation and diagnostic confidence at Philips.

Kind regards,
Kostiantyn Pysanyi